

WeatherCast 🌤️: Next-Day Rain Prediction for Maharashtra

Machine Learning Case Study 75 — Final Academic Project Report & Viva Deliverable

B.Tech Semester V | Supervised Binary Classification | 25-Year ECMWF ERA5 Dataset

🌐 Live Deployed Application:	https://weatherpredictiveml-git-geavtut3pfyypewaut2she.streamlit.app/
📁 GitHub Code Repository:	https://github.com/PIYUSH-108-SOLANKI/Weather_predictive_ml-models/best_model_phase6.joblib (Tuned Gradient Boosting Classifier)
📦 Model Artifact:	

1. PROBLEM DEFINITION & FORMULATION

Assigned Case Study 75: An environmental dataset contains historical observations, and the organization wants to identify meaningful patterns or estimate a selected future measure (With Proper Justification).

Formulated Machine Learning Task: Supervised Binary Classification to predict next-day rainfall ($\text{rain_tomorrow} \in \{0, 1\}$) across 8 meteorological stations in Maharashtra's 4 diverse agro-climatic sub-divisions (Konkan, Madhya Maharashtra, Marathwada, and Vidarbha). The target is defined as $\text{rain_tomorrow} = 1$ if next-day rainfall exceeds 0.0 mm (measurable rain), and 0 otherwise. This addresses vital agricultural irrigation scheduling in rainfed rural belts and urban drainage management during torrential coastal monsoons.

2. DATASET & PREPROCESSING PIPELINE

Data Source: 25 consecutive years (1 January 2000 – 31 December 2024, 73,056 records) of daily and hourly reanalysis data acquired from ECMWF ERA5 via the Open-Meteo Historical API across Mumbai, Ratnagiri, Pune, Nashik, Kolhapur, Solapur, Chhatrapati Sambhajinagar, and Nagpur. Calendar date-continuity was rigorously validated with 0 missing dates.

35-Feature Engineering Schema: Incorporates 1, 3, and 7-day temporal lags for precipitation, temperature, humidity, and barometric pressure; trailing 3-day and 7-day rolling aggregates (cumulative rainfall and running means); cyclical Fourier calendar projections (month_sin/cos , doy_sin/cos); and domain categoricals (IMD seasons, WMO codes, stations).

Chronological Validation: Data was split chronologically to strictly avoid temporal leakage: **Train Set:** 2000–2022 (67,208 records, 92%), **Validation Set:** 2023 (2,920 records, 4%), and **Held-out Test Set:** 2024 (2,928 records, 4%).

3. MODEL EXPERIMENTS & HELD-OUT TEST SET (2024) BENCHMARKS

Three supervised algorithm families were developed and hyperparameter-tuned on the validation split, then benchmarked on the held-out 2024 test year:

Model Architecture	Accuracy	ROC-AUC	Precision	Recall	F1-Score	Brier Score
🏆 Gradient Boosting (Tuned - Champion)	90.62%	0.9714	87.42%	84.50%	0.8593	0.0682
🌲 Random Forest (Tuned)	89.28%	0.9632	85.80%	81.90%	0.8380	0.0765
📈 Logistic Regression (Tuned)	84.15%	0.9080	78.10%	74.30%	0.7615	0.1120

Model Architecture	Accuracy	ROC-AUC	Precision	Recall	F1-Score	Brier Score
Baseline Gradient Boosting	90.18%	0.9667	86.90%	83.70%	0.8527	0.0715
Baseline Random Forest	88.75%	0.9575	85.10%	80.90%	0.8295	0.0798
Baseline Logistic Regression	83.92%	0.9051	77.80%	74.00%	0.7585	0.1145

Champion Model Selection Justification: Tuned Gradient Boosting delivered the highest test ROC-AUC (0.9714) and F1-score (0.8593), and the lowest Brier score (0.0682, indicating superior probability calibration). Its serialized artifact size (260 KB) enables sub-millisecond cloud inferences on Streamlit Cloud.

4. INTERACTIVE STREAMLIT APPLICATION

The cloud-hosted application is active at <https://weatherpredictiveml-git-geavtut3pfyypewaut2she.streamlit.app/> and provides:

- Live Real-Time Forecast:** Queries live Open-Meteo telemetry across 8 Maharashtra stations with 7-day atmospheric trend charts (temperature swings, humidity build-up, and rainfall history) and IMD advisory warnings.
- What-If Scenario Simulator:** Interactive offline sliders to stress-test model behavior under extreme monsoon downpours, scorching dry summers, and winter conditions.
- Multi-Model Comparison:** Evaluates inputs across Gradient Boosting, Random Forest, and Logistic Regression side-by-side with individual probabilities and consensus agreement.
- Decision Threshold Slider:** Customizable probability cutoff (0.10 to 0.90) demonstrating operational precision-recall trade-offs.

5. COLLEGE PORTAL SUBMISSION SUMMARY

PROJECT: WeatherCast – Next-Day Rain Prediction for Maharashtra (Case Study 75)
STUDENT: Piyush Solanki | SEMESTER: B.Tech Sem V | SUBJECT: Machine Learning Lab
LIVE APP URL: <https://weatherpredictiveml-git-geavtut3pfyypewaut2she.streamlit.app/>
GITHUB URL: https://github.com/PIYUSH-108-SOLANKI/Weather_predictive_ml-
MODEL: Tuned Gradient Boosting Pipeline (90.62% Test Accuracy, 0.9714 ROC-AUC, 0.8593 F1-Score)
DATASET: 25 Years (2000–2024, 73,056 records) ECMWF ERA5 Reanalysis across 8 Maharashtra Stations.